Lesson Plan: Probing the Invisible

From Wave Interference to Nanoparticle SAXS Analytics

- **Grade Level:** 9–12 (Physics)
- **Duration:** 3 Days (50-minute periods)
- Eman Sarhan (UIC RET Summer Program)
- **Digital Tools:** Interactive Dual-Shape SAXS Web Simulator

NGSS Alignment

- **Performance Expectation (PE):**
  - **HS-PS4-3:** Evaluate the claims, evidence, and reasoning behind the idea that electromagnetic radiation can be described either by a wave model or a particle model, and that for some situations one model is more useful than the other.
  - **HS-PS4-1:** Use mathematical representations to support a claim regarding relationships among the frequency, wavelength, and speed of waves traveling in various media.
- **Disciplinary Core Ideas (DCI):**
  - **PS4.A: Wave Properties:** Waves can add or cancel one another as they cross, depending on their relative phase (interference pattern).
  - **PS4.B: Electromagnetic Radiation:** Diffraction patterns demonstrate wave behavior.
- **Science and Engineering Practices (SEP):**
  - *Using Mathematics and Computational Thinking*
  - *Analyzing and Interpreting Data*
- **Crosscutting Concepts (CCC):**
  - *Patterns:* Observed patterns in wave scattering guide structure identification.
  - *Scale, Proportion, and Quantity*

Lesson Guiding Questions

1. How can light measure objects that are too small to see with a traditional optical microscope?
2. What connects a laser shining through a single strand of hair to high-tech X-ray scattering (SAXS) of nanoparticles?

Day 1: Young's Experiment & Hair Diffraction Lab

Objective:

Students will calculate the diameter of a human hair strand using laser diffraction, demonstrating the wave properties of light..

Materials Needed (per group):

- Red or Green Laser Pointer \lambda approx 650-532 nm)
- Standard Index Cards or Binder Clips (to mount the hair)
- A strand of human hair
- Tape, Tape Measure / Meter Stick
- White wall or cardboard screen

Procedure:

1. **Engage (10 min):** Show Thomas Young's 1801 Double-Slit experiment diagram. Ask: *"If*

light travels in straight lines, why does passing it through tiny openings create alternating light and dark bands on a screen?"

2. **Explore & Lab Execution (30 min):**
   - Students stretch a single hair across an index card hole and clip it vertically.
   - Shine the laser through the hair so the pattern projects onto a screen L=1.5 to 2.0 m away
   - **Babinet's Principle Note:** Explain that an obstacle (hair strand) produces the exact same diffraction pattern as an opening (slit) of the same width!
   - Students measure the distance y from the central bright spot to the m-th dark fringe.
3. **Mathematics / Calculation:**
   - Students apply the grating/diffraction equation
   - Solve for hair diameter d

Day 2: The Scale Jump — From Hair to Nanoparticles

Objective:

Students will understand why optical lasers cannot resolve nanometer-sized features and discover how Small-Angle X-ray Scattering (SAXS) uses shorter wavelengths to probe reciprocal space (q-space).

Core Physics Concepts:

- **Why X-rays?** Visible light wavelengths 500 nm are too long to diffract around a 10 nm nanoparticle. X-rays (lambda 0.15 nm are sized perfectly to interact with nanoscale matter
- **Reciprocal Space Reciprocity (q prop to 1/R):** Just like the hair lab where a *thinner* hair makes *wider* fringe spacing, a *smaller* nanoparticle pushes its scattering minima to *higher scattering angles* (q-space).

Day 3: Digital SAXS Simulator & Analytics Lab

Objective:

Students use an interactive web simulator to manipulate nanoparticle dimensions, evaluate experimental noise, and test the limits of automated physics algorithms.

Web Activity:

Direct students to: **https://eksarhan.github.io/saxssimulator/**

Student Inquiry Lab Sheet (3 Mini-Activities):

Activity A: The Reciprocal Size Scaling Rule

1. Set Noise to 0.00 and Shape to Sphere.
2. Set Radius R = 10 nm. Click **Calculate Size**. Note the location of the first dip (q).
3. Reduce Radius R to 5.0 nm}: . What happened to the position of the dip?
   - *Expected Observation:* The dip shifts right to a higher q value q doubles from approx 0.45 nm), proving inverse scaling!

Activity B: The Symmetry Test (Sphere vs. Cylinder)

1. Set Radius R = 10.0nm for both shapes.
2. Compare the blue curve (Sphere) and the orange curve (Cylinder)
3. *Discussion Question:* Why does the sphere have deep, sharp dips while the cylinder's dips are smoothed out?
   - *Physics Answer:* Spheres have isotropic (perfect 3D radial) symmetry, creating sharp phase cancellation. Cylinders have multi-axis lengths that smear out coherent interference fringes.

Activity C: Noise Breakdown & Machine Learning Hook

1. Keep R = 10.0 nm and gradually increase the Noise slider from 0.00 to 0.10.
2. Observe when the **Calculate Size** button begins returning errors or inaccurate values.
3. *Reflection:* In real laboratories, noise obscures physics equations. How do modern scientists decode messy data? (Introduces modern Machine Learning / Pattern Recognition algorithms).

Assessment & Evaluation

- **Formative Assessment (Lab Notebook):**
  - Hair diffraction measurement calculations from Day 1.
  - Graphing inverse relationships (d vs. fringe width y.
- **Summative Assessment (Exit Ticket / Synthesis Claim):**
  - *Prompt:* "Explain how the physics of measuring a human hair with a red laser pointers is fundamentally identical to measuring a 10 nm gold particle with an X-ray scattering machine."
  - *Rubric Criterion:* Students must reference **wave interference**, **inverse/reciprocal scaling** , and **wavelength vs. target size constraints**.